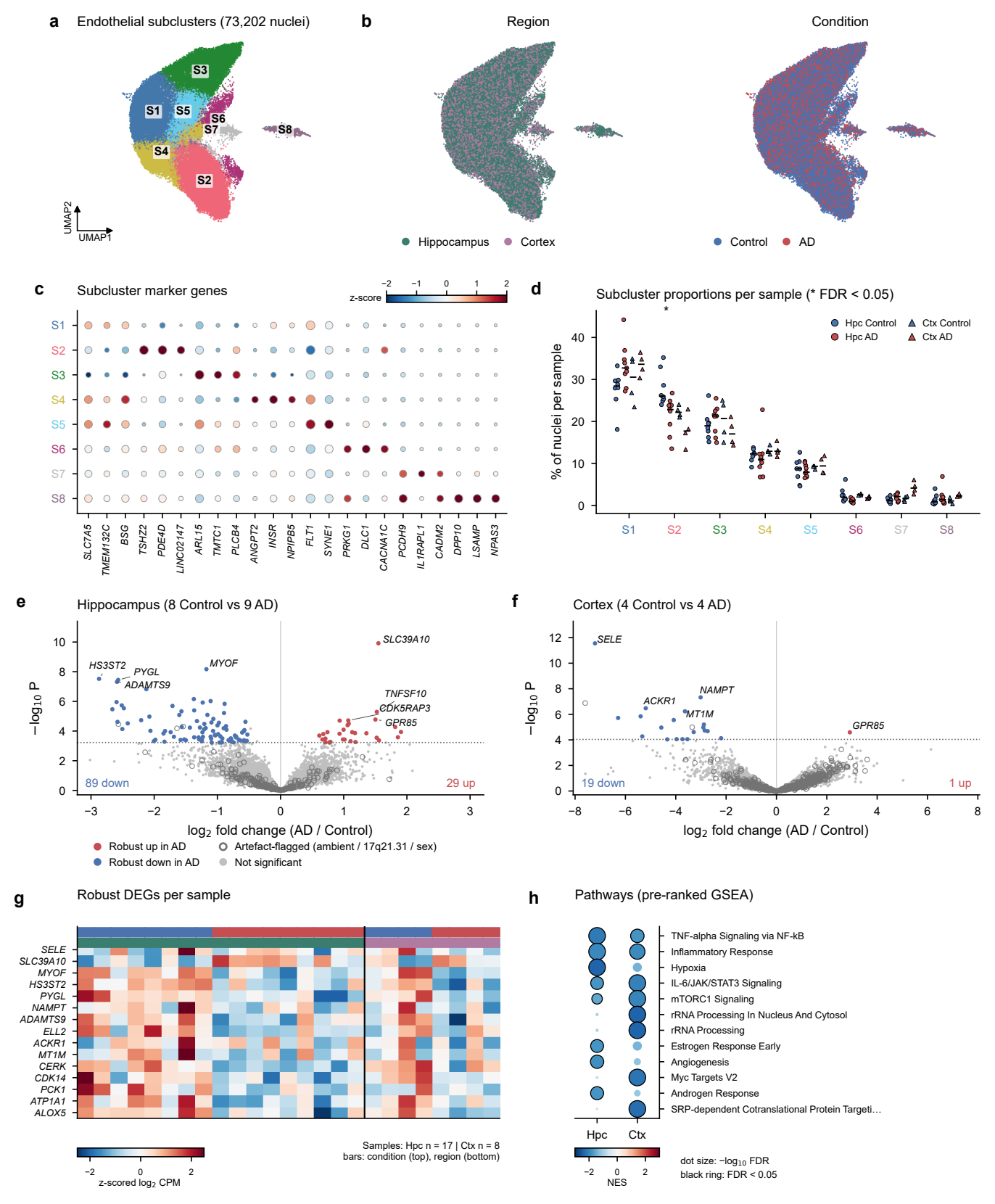

a
Endothelial subclusters (73,202 nuclei)
S1
S2
S3
S4
S5
S6
S7
S8
UMAP2
UMAP1
b
Region
Hippocampus
Cortex
Condition
Control
AD
c
Subcluster marker genes
z-score
−2 −1 0 1 2
S1 S2 S3 S4 S5 S6 S7 S8
SLC7A5 TMEM132C BSG TSHZ2 PDE4D LINC02147 ARL15 TMTC1 PLCB4 ANGPT2 INSR NPIPB5 FLT1 SYNE1 PRKG1 DLC1 CACNA1C PCDH9 IL1RAPL1 CADM2 DPP10 LSAMP NPAS3
d
Subcluster proportions per sample (* FDR < 0.05)
*
Hpc Control
Hpc AD
Ctx Control
Ctx AD
% of nuclei per sample
0 10 20 30 40
S1 S2 S3 S4 S5 S6 S7 S8
e
Hippocampus (8 Control vs 9 AD)
SLC39A10
HS3ST2
PYGL
ADAMTS9
MYOF
TNFSF10
CDK5RAP3
GPR85
89 down
29 up
−log10 P
log2 fold change (AD / Control)
Robust up in AD
Robust down in AD
Artefact-flagged (ambient / 17q21.31 / sex)
Not significant
f
Cortex (4 Control vs 4 AD)
SELE
NAMPT
ACKR1
MT1M
GPR85
19 down
1 up
−log10 P
log2 fold change (AD / Control)
g
Robust DEGs per sample
SELE
SLC39A10
MYOF
HS3ST2
PYGL
NAMPT
ADAMTS9
ELL2
ACKR1
MT1M
CERK
CDK14
PCK1
ATP1A1
ALOX5
z-scored log2 CPM
−2 0 2
Samples: Hpc n = 17 | Ctx n = 8
bars: condition (top), region (bottom)
h
Pathways (pre-ranked GSEA)
TNF-alpha Signaling via NF-kB
Inflammatory Response
Hypoxia
IL-6/JAK/STAT3 Signaling
mTORC1 Signaling
rRNA Processing In Nucleus And Cytosol
rRNA Processing
Estrogen Response Early
Angiogenesis
Myc Targets V2
Androgen Response
SRP-dependent Cotranslational Protein Targeti…
Hpc
Ctx
NES
−2 0 2
dot size: −log10 FDR
black ring: FDR < 0.05